Part A — Theory (Short Questions)

1. Nine Pillars Understanding

Why is using AI Development Agents (like Gemini CLI) for repetitive setup tasks better for your growth as a system architect?

I think using CLI based AI agent like Gemini CLI is better for repetitive task because it can develop the same code structure over and over again while i as a developer can focus more on the creativity, problem solving and management.

Explain how the Nine Pillars of AIDD help a developer grow into an M-Shaped Developer.

Nine Pillars of AIDD can help a developer become a M-Shaped developer by making it easier for the developer to create something without having too learn everything from scratch for every domain. The Nine structure of AIDD is design in a way that is beneficial for the developer if he develops a product using these pillars which can be difficult and time consuming if he develops without these AIDD Pillars.

2. Vibe Coding vs Specification-Driven Development

Why does Vibe Coding usually create problems after one week?

Vibe Coding usually creates problems because when the developer is developing something they don't think about scaling or maintaining it later they just work on the spot and live test it until it's finished. Later when they want to add a feature to the product it becomes problematic because the code structure is messy and not maintainable.

How would Specification-Driven Development prevent those problems?

Specification-Driven Development solve these problems before they even occur meaning that the structure of the development is decided before starting coding and it is designed in a detailed structured way using markdown as the ultimate source and structure for the product so that when using the first pillar which is AI agents uses this markdown it sets some rules while development so it can maintain the code if the ultimate source which is our markdown changes or modified.

3. Architecture Thinking

How does architecture-first thinking change the role of a developer in AIDD?

architechture-first thinking rule changes the role of the developer to think like a architech for it's product and it forces the developer to create a detailed structure before aimelessly just starts to code without even thinking

Explain why developers must think in layers and systems instead of raw code.

as i explain earlier the developer have to carefully thinks like an architech and think in layers instead of just developing with raw code when the developer thinks in layer it gives them the advantage of maintanence

Part B — Practical Task (Screenshot Required)

Task:

Using any AI CLI tool, generate a 1-paragraph specification for an email validation function.

Requirements:

Must contain “@”

Must contain a valid domain (e.g., .com, .org)

Should return clear error messages

Specification generated by Gemini CLI:

The email validation function shall verify that a given string adheres to standard email formatting rules. This includes ensuring the presence of a single "@" symbol separating the local part from the domain, and that the domain itself contains a top-level domain (TLD) with at least two characters (e.g., ".com", ".org", ".net"). The function must return distinct, human-readable error messages for each type of validation failure, such as "Email must contain @" for a missing '@' symbol or "Invalid domain extension" for an improperly formatted or missing TLD.

Part C — Multiple Choice Questions

1.B

2.B

3.B

4.B

5.C